Unbiasedness of the Difference-in-Means Estimator for the SATE and the PATE*

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This guide proves that the Difference-in-Means estimator is unbiased for the Sample Average Treatment Effect (SATE), under both complete and simple random assignment, and, when the study sample is itself a random draw from a superpopulation, that the estimator is also unbiased for the Population Average Treatment Effect (PATE). This guide states the framework only as far as its proofs require and shares that framework with three companion guides: "Notation and Setup for Design-Based Causal Inference" develops the notation in full, and "Variance of the Difference-in-Means Estimator for the SATE and Estimation of That Variance" and "Variance of the Difference-in-Means Estimator for the PATE and Estimation of That Variance" take up the estimator's variance.

1 Estimation of Sample Average Treatment Effect (SATE)

1.1 Setup

This section states the framework only as far as the proofs require; the companion guide "Notation and Setup for Design-Based Causal Inference" develops each object in full, with the reasoning behind it.

Let the index $i \in \{1, \ldots, n\}$ run over the n units of a finite sample, $\mathcal{S}_n$, of which $n_T \geq 2$ receive treatment and $n_C = n - n_T \geq 2$ receive control, so that $n \geq 4$ (these bounds matter only for

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the variance estimator in the companion guides; the setup guide explains them). The indicator $Z_i \in \{0,1\}$ records whether unit i receives treatment ($Z_i = 1$) or control ($Z_i = 0$), and the random assignment vector $\boldsymbol{Z} = \left[Z_1, \ldots, Z_n\right]^\top$ collects the indicators.

Every assignment vector determines how many units it sends to treatment: Write $n_T(\boldsymbol{z}) \coloneqq \sum_{i=1}^n z_i$ for this count and $n_C(\boldsymbol{z}) \coloneqq n - n_T(\boldsymbol{z})$ for its complement, so that $n_T(\boldsymbol{Z})$ is a random variable. Under *complete random assignment* the design fixes the count: The support of $\boldsymbol{Z}$ is $\Omega = \{\boldsymbol{z} \in \{0,1\}^n : n_T(\boldsymbol{z}) = n_T\}$, on which $n_T(\boldsymbol{Z})$ equals the design constant n_T with probability one, and $\boldsymbol{Z}$ is distributed uniformly on Ω, whose cardinality[1] is $|\Omega| = \binom{n}{n_T}$.

Under *simple random assignment* no constraint fixes the count, and $n_T(\boldsymbol{Z})$ is a nondegenerate random variable; Section 1.3 treats that case by conditioning on the event $\{n_T(\boldsymbol{Z}) = n_T\}$.

Under the Stable Unit Treatment Value Assumption (SUTVA) (Cox, 1958; Rubin, 1980, 1986), unit i's potential outcome depends on the assignment vector only through unit i's own coordinate, so each unit carries two *fixed* potential outcomes, $y_i(1)$ and $y_i(0)$, with individual effect $\tau_i \coloneqq y_i(1) - y_i(0)$. The observed outcome is $Y_i \coloneqq y_i(Z_i) = Z_i y_i(1) + (1 - Z_i) y_i(0)$, random only through Z_i. The setup guide develops the potential outcomes schedule, states SUTVA precisely, and walks through the collapse from $y_i(\boldsymbol{Z})$ to $y_i(1)$ and $y_i(0)$ in numbered steps.

Write $\bar{y}(1) \coloneqq n^{-1} \sum_{i=1}^n y_i(1)$ and $\bar{y}(0) \coloneqq n^{-1} \sum_{i=1}^n y_i(0)$ for the finite population means of the treatment and control potential outcomes; both are fixed quantities. The target of interest is the Sample Average Treatment Effect (SATE),

$$\tau_{\text{SATE}} \coloneqq n^{-1} \sum_{i=1}^n \tau_i = n^{-1} \sum_{i=1}^n \big(y_i(1) - y_i(0)\big) = \bar{y}(1) - \bar{y}(0),$$

which is likewise a fixed (though unknown) quantity. Define the Difference-in-Means estimator of τ_{SATE} as

$$\hat{\tau} \coloneqq \frac{\sum_{i=1}^n Z_i Y_i}{\sum_{i=1}^n Z_i} - \frac{\sum_{i=1}^n (1 - Z_i)\, Y_i}{\sum_{i=1}^n (1 - Z_i)}. \tag{1}$$

Unlike the estimand τ_{SATE}, the estimator $\hat{\tau}$ is a *random variable*, since $\hat{\tau}$ is a function of the random assignment vector $\boldsymbol{Z}$ and inherits all randomness from $\boldsymbol{Z}$. The denominators of the ratio form are the counts $n_T(\boldsymbol{Z})$ and $n_C(\boldsymbol{Z})$; under complete random assignment the design fixes them at n_T and

[1] For an arbitrary set W, let $|W|$ denote the cardinality of (i.e., the number of elements in) the set W.

n_C, so we can equivalently write $\hat{\tau}$ as

$$\hat{\tau} = \frac{1}{n_T} \sum_{i=1}^{n} Z_i Y_i - \frac{1}{n_C} \sum_{i=1}^{n} (1 - Z_i)\, Y_i,$$

which is the form that the companion guides on the estimator's variance use. For the expectation of $\hat{\tau}$ in Equation (1), we write $\mathrm{E}_\Omega[\cdot]$ to emphasize that the expectation runs over only the randomness of the assignment process, i.e., over the distribution of $\boldsymbol{Z}$ on Ω.

1.2 Estimation under Complete Random Assignment

Lemma 1. *Under complete, uniform random assignment in which the design sends n_T of n total units to treatment, $\mathrm{E}_\Omega[Z_i] = \frac{n_T}{n}$ for every unit $i \in \{1, \ldots, n\}$.*

Proof. We will complete this proof in two steps: We will show that (1) the proportion of assignments in which unit i is in the treatment condition is $\frac{n_T}{n}$ and (2) under uniform, random assignment, the probability that $Z_i = 1$ is equal to this proportion $\frac{n_T}{n}$.

Step 1: The number of ways to choose a subset of n_T treated units from a fixed population of n units is

(2)
$$\binom{n}{n_T} = \frac{n!}{(n - n_T)! n_T!} = \frac{n!}{n_C! n_T!},$$

where $n_C = n - n_T$ is the number of units in the control condition.

Given that an arbitrary unit i is in the treatment condition and only n_T total units can be in the treatment condition, there are $\binom{n-1}{n_T-1}$ ways in which $n_T - 1$ other units could be in the treatment condition. Hence, the number of assignments in which unit i is treated and $n_T - 1$ other units are treated is:

(3)
$$\binom{n-1}{n_T - 1} = \frac{(n-1)!}{\big((n-1) - (n_T - 1)\big)!\, (n_T - 1)!}$$

To get the proportion of assignments in which unit i is treated, we need to divide (3) by (2):

$$
(4) \qquad \frac{\binom{n-1}{n_T-1}}{\binom{n}{n_T}} = \frac{\left(\dfrac{(n-1)!}{((n-1)-(n_T-1))!\,(n_T-1)!}\right)}{\left(\dfrac{n!}{n_C!n_T!}\right)}
$$

Now notice that:

$$
\begin{aligned}
(n-1)-(n_T-1) &= n-1-n_T+1 \\
&= n-n_T \\
&= n_C
\end{aligned}
$$

We can therefore substitute n_C for $(n-1)-(n_T-1)$ in (4), which gives us:

$$
(5) \qquad \frac{\left(\dfrac{(n-1)!}{n_C!\,(n_T-1)!}\right)}{\left(\dfrac{n!}{n_C!n_T!}\right)}
$$

Now we can simply manipulate (5) and cancel terms until we are left with $\frac{n_T}{n}$:

$$
\begin{aligned}
&= \left(\frac{(n-1)!}{n_C!\,(n_T-1)!}\right)\left(\frac{n_C!n_T!}{n!}\right) \\
&= \left(\frac{(n-1)\,(n-2)\ldots 1}{n_C\,(n_C-1)\ldots 1\,(n_T-1)\ldots 1}\right)\left(\frac{n_C\,(n_C-1)\ldots 1 n_T\,(n_T-1)\ldots 1}{n\,(n-1)\ldots 1}\right) \\
&= \frac{\overbrace{(n-1)\,(n-2)\ldots 2}^{\text{(a)}}\,\overbrace{n_C\,(n_C-1)\ldots 2}^{\text{(b)}}\,n_T\,\overbrace{(n_T-1)\ldots 2}^{\text{(c)}}}{\underbrace{n_C\,(n_C-1)\ldots 2}_{\text{(b)}}\,\underbrace{(n_T-1)\ldots 2}_{\text{(c)}}\,n\,\underbrace{(n-1)\ldots 2}_{\text{(a)}}}
\end{aligned}
$$

Each braced group in the numerator has an identical partner in the denominator: (a) with (a), (b) with (b), and (c) with (c). All three pairs cancel, which leaves us with $\frac{n_T}{n}$. Therefore, unit i is in the treatment condition in exactly the fraction $\frac{n_T}{n}$ of all possible assignments.

Step 2: The probability that unit i is treated is the sum of the probabilities of the assignments in which unit i is treated. Under uniform random assignment, each assignment has probability $\frac{1}{|\Omega|}$. Thus,

$$\begin{aligned}\left(\frac{1}{|\Omega|}\right)\left(\frac{n_T}{n}\right)|\Omega| &= \left(\frac{1}{|\Omega|}\right)\frac{n_T\,|\Omega|}{n} \\ &= \frac{n_T\,|\Omega|}{|\Omega|\,n} \\ &= \frac{n_T}{n}\end{aligned}$$

Since $\Pr_\Omega(Z_i = 1) = \frac{n_T}{n}$ for every unit $i \in \{1, \ldots, n\}$, the expected value of $Z_i \in \{0, 1\}$ is $\mathrm{E}_\Omega[Z_i] = 1(\frac{n_T}{n}) + 0(1 - \frac{n_T}{n}) = \frac{n_T}{n}$. □

Proposition 1. *Under complete, uniform random assignment,* $\mathrm{E}_\Omega[\hat{\tau}] = \tau_{SATE}$.

Proof. First, the linearity of expectations implies that

$$\begin{aligned}\mathrm{E}_\Omega\left[\hat{\tau}\right] &= \mathrm{E}_\Omega\left[\frac{\sum_{i=1}^{n} Z_i Y_i}{\sum_{i=1}^{n} Z_i} - \frac{\sum_{i=1}^{n}\left(1 - Z_i\right) Y_i}{\sum_{i=1}^{n}\left(1 - Z_i\right)}\right] \\ &= \mathrm{E}_\Omega\left[\frac{\sum_{i=1}^{n} Z_i Y_i}{\sum_{i=1}^{n} Z_i}\right] - \mathrm{E}_\Omega\left[\frac{\sum_{i=1}^{n}\left(1 - Z_i\right) Y_i}{\sum_{i=1}^{n}\left(1 - Z_i\right)}\right]\end{aligned}$$

and, since complete random assignment fixes the counts at n_T and n_C,

$$\mathrm{E}_\Omega\left[\hat{\tau}\right] = \frac{1}{n_T}\,\mathrm{E}_\Omega\left[\sum_{i=1}^{n} Z_i Y_i\right] - \frac{1}{n_C}\,\mathrm{E}_\Omega\left[\sum_{i=1}^{n}\left(1 - Z_i\right) Y_i\right]$$

Since, under SUTVA, the observed outcomes for treated units are equal to those units' treatment potential outcomes, we can substitute $Z_i y_i(1)$ for $Z_i Y_i$. Analogously, we can substitute $(1 - Z_i) y_i(0)$ for $(1 - Z_i)Y_i$. That is, with $Y_i = Z_i y_i(1) + (1 - Z_i) y_i(0)$, it follows that

$$\begin{gathered}Z_i Y_i = Z_i\left(Z_i y_i(1) + (1 - Z_i) y_i(0)\right) = Z_i y_i(1) \text{ and} \\ (1 - Z_i) Y_i = (1 - Z_i)\left(Z_i y_i(1) + (1 - Z_i) y_i(0)\right) = (1 - Z_i) y_i(0),\end{gathered}$$

which leaves us with

$$
\begin{aligned}
\mathrm{E}_\Omega\left[\hat{\tau}\right] &= \frac{1}{n_T}\,\mathrm{E}_\Omega\left[\sum_{i=1}^{n} Z_i y_i(1)\right] - \frac{1}{n_C}\,\mathrm{E}_\Omega\left[\sum_{i=1}^{n} \left(1 - Z_i\right) y_i(0)\right] \\
&= \left(\frac{1}{n_T}\right)\mathrm{E}_\Omega\left[Z_1 y_1(1) + \cdots + Z_n y_n(1)\right] - \left(\frac{1}{n_C}\right)\mathrm{E}_\Omega\left[\left(1 - Z_1\right) y_1(0) + \cdots + \left(1 - Z_n\right) y_n(0)\right] \\
&= \left(\frac{1}{n_T}\right)\Big(\mathrm{E}_\Omega\left[Z_1 y_1(1)\right] + \cdots + \mathrm{E}_\Omega\left[Z_n y_n(1)\right]\Big) \\
&\quad - \left(\frac{1}{n_C}\right)\Big(\mathrm{E}_\Omega\left[\left(1 - Z_1\right) y_1(0)\right] + \cdots + \mathrm{E}_\Omega\left[\left(1 - Z_n\right) y_n(0)\right]\Big) \\
&= \left(\frac{1}{n_T}\right)\left(y_1(1)\,\mathrm{E}_\Omega\left[Z_1\right] + \cdots + y_n(1)\,\mathrm{E}_\Omega\left[Z_n\right]\right) \\
&\quad - \left(\frac{1}{n_C}\right)\Big(y_1(0)\,\mathrm{E}_\Omega\left[\left(1 - Z_1\right)\right] + \cdots + y_n(0)\,\mathrm{E}_\Omega\left[\left(1 - Z_n\right)\right]\Big)
\end{aligned}
$$

By Lemma 1, $\mathrm{E}_\Omega[Z_i] = (\frac{n_T}{n})$ for all $i \in \{1, \ldots, n\}$, which implies that $\mathrm{E}_\Omega[(1 - Z_i)] = 1 - (\frac{n_T}{n}) = (\frac{n_C}{n})$ for all $i \in \{1, \ldots, n\}$. Hence, we can substitute $(\frac{n_T}{n})$ for $\mathrm{E}_\Omega[Z_i]$ and $(\frac{n_C}{n})$ for $\mathrm{E}_\Omega[1 - Z_i]$, respectively, which then yields

$$
\begin{aligned}
&= \left(\frac{1}{n_T}\right)\left(\frac{n_T}{n}\right)\big(y_1(1) + \cdots + y_n(1)\big) - \left(\frac{1}{n_C}\right)\left(\frac{n_C}{n}\right)\big(y_1(0) + \cdots + y_n(0)\big) \\
&= \left(\frac{1}{n}\right)\big(y_1(1) + \cdots + y_n(1)\big) - \left(\frac{1}{n}\right)\big(y_1(0) + \cdots + y_n(0)\big) \\
&= \frac{\big(y_1(1) + \cdots + y_n(1)\big)}{n} - \frac{\big(y_1(0) + \cdots + y_n(0)\big)}{n} \\
&= \bar{y}(1) - \bar{y}(0) \\
&= \tau_{\mathrm{SATE}}.
\end{aligned}
$$

□

1.3 Estimation under Simple Random Assignment

Under simple random assignment, the number of experimental units, n, remains fixed, but the arm counts are genuinely random: No design constraint pins $n_T(\boldsymbol{Z}) = \sum_{i=1}^{n} Z_i$ to a single value, and $n_C(\boldsymbol{Z}) = n - n_T(\boldsymbol{Z})$ inherits its randomness. We need no new notation. The count is random for the same reason $\hat{\tau}$ is, namely that the count is a function of the random vector $\boldsymbol{Z}$, and we mark that dependence by displaying the argument. When we condition on the event $\{n_T(\boldsymbol{Z}) = n_T\}$, the plain n_T denotes the realized value, fixed from that point on, exactly as z_i denotes the realized value of Z_i.

We take the support of $n_T(\boldsymbol{Z})$ to be $\{1, \ldots, n-1\}$, so that neither arm can be empty (an empty arm would leave the estimator $\hat{\tau}$ undefined). The set of possible assignments is therefore $\Omega = \{\boldsymbol{z} \in \{0,1\}^n : 0 < n_T(\boldsymbol{z}) < n\}$, which contains $2^n - 2$ elements; $\boldsymbol{Z}$ follows the distribution of n independent Bernoulli draws conditioned on the event $\{0 < n_T(\boldsymbol{Z}) < n\}$. Whenever the proof below conditions on the event $\{n_T(\boldsymbol{Z}) = n_T\}$, the resulting conditional distribution is uniform on $\{\boldsymbol{z} : n_T(\boldsymbol{z}) = n_T\}$, which is exactly the complete random assignment distribution; we write the conditioning explicitly rather than changing the symbol Ω.

In the proof that follows, we will draw upon the Law of Iterated Expectations: For two random variables X and W defined on the same probability space, $\mathrm{E}[X] = \mathrm{E}[\mathrm{E}[X \mid W]]$. In our finite setting the conditioning variable is the count $n_T(\boldsymbol{Z})$, and the law takes the form of the weighted sum in Equation (6) below.

Proposition 2. *Under simple, uniform random assignment,* $\mathrm{E}_\Omega[\hat{\tau}] = \tau_{SATE}$.

Proof. By the law of iterated expectations, we can decompose the expected value of the Difference-in-Means estimator, $\hat{\tau}$, by conditioning on the realized number of treated units, $n_T(\boldsymbol{Z})$:

$$\begin{aligned} \mathrm{E}_\Omega\left[\hat{\tau}\right] = &\ \mathrm{E}_\Omega\left[\hat{\tau} \mid n_T(\boldsymbol{Z}) = 1\right] \Pr_\Omega\left(n_T(\boldsymbol{Z}) = 1\right) \\ &+ \cdots + \mathrm{E}_\Omega\left[\hat{\tau} \mid n_T(\boldsymbol{Z}) = n-1\right] \Pr_\Omega\left(n_T(\boldsymbol{Z}) = n-1\right). \end{aligned} \tag{6}$$

Conditional on the event $\{n_T(\boldsymbol{Z}) = n_T\}$, simple random assignment reduces to complete random assignment with n_T treated units, so Proposition 1 applies. By that proposition, the expected value of the estimator conditional on any realized number of treated units $n_T \in \{1, \ldots, n-1\}$ is equal to τ_{SATE}. Hence, we can rewrite Equation (6) as

$$\mathrm{E}_\Omega\left[\hat{\tau}\right] = \tau_{\text{SATE}} \Pr_\Omega\left(n_T(\boldsymbol{Z}) = 1\right) + \cdots + \tau_{\text{SATE}} \Pr_\Omega\left(n_T(\boldsymbol{Z}) = n-1\right),$$

which we can rewrite as

$$\mathrm{E}_\Omega\left[\hat{\tau}\right] = \tau_{\text{SATE}}\Big[\Pr_\Omega\left(n_T(\boldsymbol{Z}) = 1\right) + \cdots + \Pr_\Omega\left(n_T(\boldsymbol{Z}) = n-1\right)\Big].$$

Finally, note that, since $n_T(\boldsymbol{Z})$ takes values in $\{1, \ldots, n-1\}$ with probability one, the second and third axioms of probability imply that $\Big[\Pr_\Omega(n_T(\boldsymbol{Z}) = 1) + \cdots + \Pr_\Omega(n_T(\boldsymbol{Z}) = n-1)\Big] = 1$. Hence,

$$\begin{aligned} \mathrm{E}_\Omega\left[\hat{\tau}\right] &= \tau_{\text{SATE}}\Big[1\Big] \\ \mathrm{E}_\Omega\left[\hat{\tau}\right] &= \tau_{\text{SATE}}, \end{aligned}$$

which proves the proposition. $\square$

2 Estimation of the Population Average Treatment Effect (PATE)

2.1 Setup

So far we have treated the n units of $\mathcal{S}_n$ as the entire population of interest and shown that $\hat{\tau}$ is unbiased for the SATE. Suppose now that these n units are themselves a random sample from a larger superpopulation, and ask whether $\hat{\tau}$ is unbiased for the average treatment effect in that superpopulation. (The companion guides develop the full superpopulation framework and the variance of $\hat{\tau}$ about the PATE; here we introduce only what unbiasedness requires.)

Consider a superpopulation $\mathcal{P}_N$ of size $N \geq n$, indexed by $i \in \{1, \ldots, N\}$, in which every unit has fixed potential outcomes $y_i(1)$ and $y_i(0)$ and a fixed individual effect $\tau_i = y_i(1) - y_i(0)$. Let simple random sampling draw n of these N units into the experimental sample $\mathcal{S}_n$; the indicator $R_i \in \{0, 1\}$ records the outcome, taking $R_i = 1$ if unit i enters the sample and $R_i = 0$ otherwise. Write $n(\boldsymbol{r}) \coloneqq \sum_{i=1}^{N} r_i$ for the number of units a sampling vector selects, so that the sampling vector $\boldsymbol{R} = \left[R_1, \ldots, R_N\right]^\top$ is uniform on $\Pi = \{\boldsymbol{r} \in \{0, 1\}^N : n(\boldsymbol{r}) = n\}$, the same level set construction that defines Ω. Among the n sampled units, complete random assignment sends n_T to treatment and n_C to control, exactly as before. The estimand is the Population Average Treatment Effect,

$$\tau_{\text{PATE}} \coloneqq \frac{1}{N} \sum_{i=1}^{N} \tau_i,$$

a fixed (though unknown) feature of the superpopulation, and the Difference-in-Means estimator becomes

$$\hat{\tau} \coloneqq \frac{1}{n_T} \sum_{i=1}^{N} R_i Z_i Y_i - \frac{1}{n_C} \sum_{i=1}^{N} R_i \left(1 - Z_i\right) Y_i, \tag{7}$$

in which $R_i Z_i = 1$ only for units that are both sampled and treated, $R_i(1 - Z_i) = 1$ only for units that are both sampled and control, and unsampled units ($R_i = 0$) contribute nothing. There are now *two* sources of randomness, the sampling indicators $\{R_i\}_{i=1}^{N}$ and the assignment indicators $\{Z_i\}_{i=1}^{n}$. We write $\mathrm{E}[\cdot]$ for the expectation over both, $\mathrm{E}_\Pi[\cdot]$ for the expectation over the sampling process, and $\mathrm{E}_\Omega[\cdot]$ for the expectation over the assignment process conditional on the realized sample.

Proposition 3. *Under simple random sampling of n units from $\mathcal{P}_N$ and complete random assign-*

ment among the sampled units, $\mathrm{E}[\hat{\tau}] = \tau_{PATE}$.

Proof. We take the expectation over both stages of randomness by iterating, first over the assignment (conditional on the realized sample) and then over the sample,

$$\mathrm{E}\left[\hat{\tau}\right] = \mathrm{E}_{\Pi}\left[\mathrm{E}_{\Omega}\left[\hat{\tau}\right]\right].$$

Conditional on any realized sample, the n sampled units undergo complete random assignment, so Proposition 1 applies *within* the sample and gives $\mathrm{E}_{\Omega}[\hat{\tau}] = \tau_{\mathrm{SATE}}(\boldsymbol{R})$, where $\tau_{\mathrm{SATE}}(\boldsymbol{R}) \coloneqq \frac{1}{n}\sum_{i=1}^{N} R_i\tau_i$ is the average effect among the sampled units. Within a fixed sample, as in the earlier sections, this average is a constant; here it is a random variable, because the sampling vector $\boldsymbol{R}$ determines which units enter the average, and we display the argument to mark that dependence. Substituting,

$$\mathrm{E}\left[\hat{\tau}\right] = \mathrm{E}_{\Pi}\left[\tau_{\mathrm{SATE}}(\boldsymbol{R})\right] = \mathrm{E}_{\Pi}\left[\frac{1}{n}\sum_{i=1}^{N} R_i\tau_i\right].$$

Because the individual effects τ_i are fixed and only the sampling indicators R_i are random, the linearity of expectations gives

$$\mathrm{E}\left[\hat{\tau}\right] = \frac{1}{n}\sum_{i=1}^{N} \tau_i \ \mathrm{E}_{\Pi}\left[R_i\right].$$

It remains to find $\mathrm{E}_{\Pi}[R_i]$. Under simple random sampling of n units from N, the event $\{R_i = 1\}$ plays exactly the role that $\{Z_i = 1\}$ played under complete random assignment of n_T units from n. The proportion of samples that include unit i is $\frac{n}{N}$, and under uniform sampling the probability of inclusion equals that proportion. The argument of Lemma 1, with N and n in place of n and n_T, therefore gives $\mathrm{E}_{\Pi}[R_i] = \frac{n}{N}$ for every $i \in \{1, \ldots, N\}$. Substituting,

$$\mathrm{E}\left[\hat{\tau}\right] = \frac{1}{n}\sum_{i=1}^{N} \tau_i\left(\frac{n}{N}\right) = \frac{1}{N}\sum_{i=1}^{N} \tau_i = \tau_{\mathrm{PATE}},$$

which proves the proposition. □

This argument reuses the earlier results rather than repeating them. The complete random assignment proposition handles the inner expectation, and the counting argument of Lemma 1 handles the outer expectation, now applied to sampling n of N units instead of assigning n_T of n. The result also makes precise the sense in which the Difference-in-Means estimator answers two questions at once. With respect to the assignment alone, the estimator is unbiased for the SATE, the average effect among the units actually studied. Once we also average over which units are sampled, the estimator is unbiased for the PATE, the average effect in the superpopulation.

References

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